

HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • Stamp 1G — Full-Time Work Eligible

PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Developed and debugged **C++ firmware** for embedded systems, implementing **BLE and Wi-Fi communication stacks** on **ESP32 microcontrollers**, managing memory constraints in resource-limited environments.
- Designed and implemented **Modbus RTU over RS-485 drivers** for industrial energy meters, enabling reliable data exchange with upstream systems and ensuring accurate **register map** interpretation.
- Authored **Python scripts** for automating the testing and provisioning of connected home devices, integrating with **serial and USB interfaces** on Linux development stations.
- Contributed to the **full firmware lifecycle** for smart home devices, from initial architecture and protocol integration through to production deployment and maintenance.
- Collaborated with hardware engineers to define and integrate **embedded application logic**, ensuring seamless interaction between device peripherals and cloud-based IoT infrastructure.
- Implemented **security fundamentals** for IoT devices, including **TLS authentication** and secure coding practices to protect device integrity and data transmission.
- Managed firmware releases using **Git-based version control**, writing **unit and integration tests**, and maintaining clear technical documentation for team collaboration and code quality.

SKILLS

Embedded C/C++ – Embedded C++, Bare-metal, RTOS, FreeRTOS, Memory Management, Object-Oriented Design

Microcontrollers – ESP32, ESP-IDF, STM32, Nordic nRF5x, RP2040, MCU

Wireless Protocols – Bluetooth, BLE, Wi-Fi, Matter, Thread

Wired Protocols & IoT – Modbus RTU, RS-485, I2C, UART, MQTT, REST APIs, WebSockets, Protobuf, AWS IoT Core, Azure IoT Hub

Linux & Scripting – Linux, Shell Scripting, Python, Cross-compilation

Security – Public-key cryptography, TLS Authentication, Secure Boot, Secure Coding

Development Practices – Git, Unit Testing, Integration Testing, Documentation, Jenkins, Docker, CI/CD

PROJECTS

Smart Home Device Connectivity Firmware (HCL Technologies)

- Developed firmware for a smart home hub using **ESP32**, implementing **Wi-Fi provisioning and BLE** communication for device discovery and configuration.
- Integrated **Modbus TCP** drivers to interface with smart energy meters, parsing sensor data and relaying it to a cloud backend via **MQTT**.
- Authored shell scripts on **Linux** to automate firmware flashing and initial device setup during manufacturing.

Embedded Systems Protocol Integration Lab (MSc Project)

- Designed and implemented a custom communication protocol over **UART** for inter-device messaging on an **STM32** microcontroller.
- Developed a Python-based test harness to simulate various network conditions and validate the reliability of the embedded communication stack.
- Utilized **RTOS** concepts to manage concurrent data streams and ensure timely processing of sensor inputs.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India

2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)